

Non-LAMP baseline verification report

A deterministic, artifact-backed assessment of FLattacks V2, DAGER-Pythia-adapted and a FILM-inspired Pythia support-constrained LM beam without reordering.

Verdict: the reported development numbers are reproducible and numerically correct

Scope boundary. This verifies eight seed-0 development aggregates. It does not convert them into repeated, held-out, publication-grade estimates.

CELLS

8 prescribed non-LAMP configurations

METHOD EXECUTIONS

24/24 successful

EVIDENCE BUNDLE

SHA - 256

cda2b3c36d26d7e5e0063956d1ce39661fa3ebf6a131607ef5f442
fec305546b

VERIFIED CODE

Repository HEAD 106dd2e; non-LAMP behavior
frozen from f552779

Contents and headline conclusions

1. What was run
2. How correctness was verified
3. Autonomous results
4. V2 truth-assisted diagnostic
5. Support, rank and resources
6. Qualitative reconstructions
7. Interpretation by method
8. Limits and remaining blockers
9. Provenance and reproducibility

Numerical conclusion. Across duplicate runs of all eight cells, 32/32 canonical/method-view gradient tensor files were byte-identical, all 24 DAGER/FILM-inspired/V2 semantic prediction payloads were identical, and all 24 metric dictionaries and assignment lists were exactly equal.

Strongest autonomous result

DAGER on the singleton cell: set RL 0.657. Its best candidate is nearly exact (matched RL 0.986), but an extra repetitive output reduces set precision.

Most persistent signal

FILM and V2 retain token-support F1 1.000 across every completed cell, while autonomous ordering remains weak.

1. What was run

The victim is a frozen pretrained causal language model. Each CSV row is one private sequence. The trusted victim concatenates $M=N \times B$ rows, right-truncates/right-pads to S , computes shifted next-token cross entropy, masks padding, and performs one global active-target-mean backward pass.

MODEL	N	B	M	S	ACTIVE TARGETS	VICTIM LOSS
Pythia-160m	1	1	1	64	35	4.7598
Pythia-160m	2	2	4	64	189	3.3237
Pythia-160m	4	2	8	128	388	3.3617
Pythia-160m	16	2	32	64	1462	3.7245
Pythia-410m	1	1	1	64	35	4.3255
Pythia-410m	4	1	4	128	206	2.8974
Pythia-410m	4	2	8	128	388	3.0176
Pythia-410m	8	2	16	128	788	3.2419

N = nominal clients; **B** = sequences per nominal client; **M=N×B** = total private sequences; **S** = public maximum tokenized sequence length. Under this primary global-mean victim, N and B affect the gradient only through M and selected rows; the client partition itself is

not a causal variable.

V2 Exact preserved mentor source, checksum-bound. Autonomous output is a fragment set. Truth-assisted assembly is separate.	DAGER GPT-NeoX fused-QKV adaptation using gradient-derived subspace/rank evidence. Extra outputs are penalized.	FILM Support-constrained Pythia adaptation without the planned candidate-gradient-norm reorder stage.	Excluded LAMP is absent. No model, tokenizer or prior was trained or fine-tuned.
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2. How correctness was verified

1. Replayed all four Pythia-160m and all four Pythia-410m non-LAMP cells with explicit N, B, M and S.
2. Verified every run exited successfully and produced canonical gradient, restricted method views, prediction JSON, receipt JSON and evaluation report.
3. Compared duplicate-run tensor bytes directly: 32/32 canonical/DAGER/FILM/V2 gradient files matched exactly.
4. Compared semantic prediction payloads: 24/24 token-ID payloads matched exactly across all eight cells.
5. Compared complete evaluator metric dictionaries and assignment lists: 24/24 matched exactly.
6. Hashed prediction and receipt files directly: all 24 files of each type matched their claimed SHA-256 identifiers.
7. Recomputed V2 assisted R1/R2/RL from frozen fragments with exact preserved `assign_and_score()` and `repository_rouge_n/_rouge_l`.
8. Reconciled all 32 displayed R1/R2/RL triplets against the verified JSON in exact order.
9. Re-ran quality gates: 57 tests passed; Ruff, formatter and mypy passed on `src/tests/scripts`.
10. Rechecked immutable V2 and development-data SHA-256 values.
11. Reconciled two independent read-only reviews: numerical replay evidence and scientific-validity/overclaiming risk.

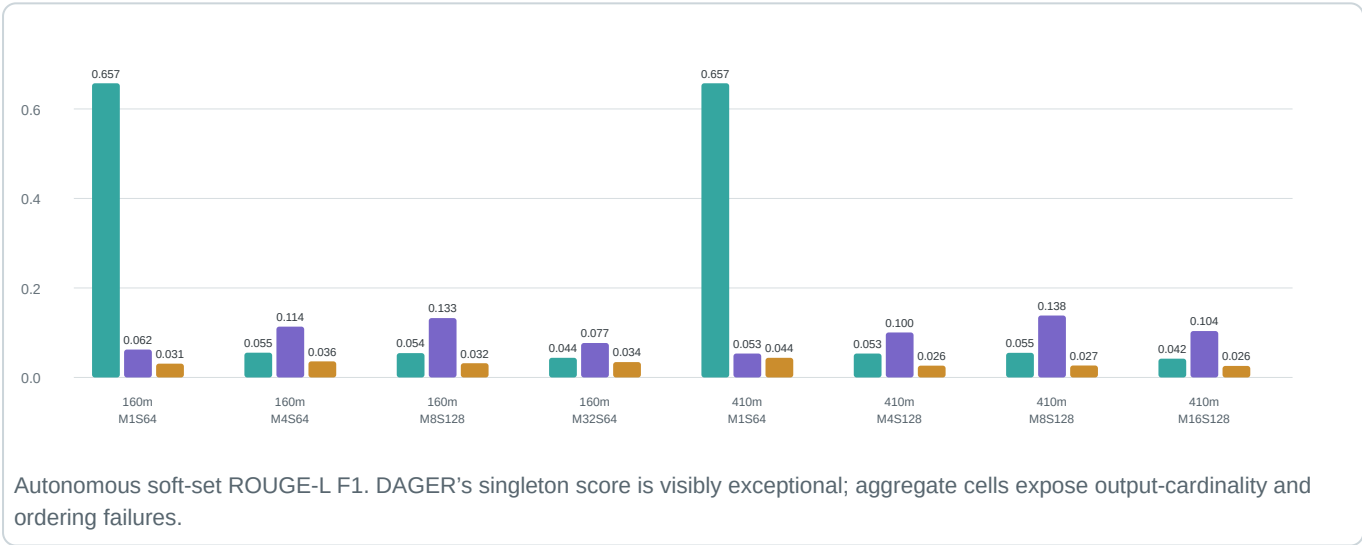
Why some envelope hashes changed. Explicit N/B replay IDs differ from earlier M-only exploratory IDs, so run-ID-bound manifests, receipts and report files have different hashes. The gradient tensor bytes and semantic predictions are identical; this is metadata rebinding, not a result difference.

LAYER	EVIDENCE	OUTCOME
Victim	Loss, target count, canonical tensor bytes	Exact match
Method input	DAGER/FILM/V2 restricted-view tensor bytes	Exact match
Attack output	Token IDs and output cardinality	Exact match
Evaluation	Full metric dictionaries	Exact match
Oracle diagnostic	Reassembled V2 fragments, repository metric functions	Independent recomputation match

3. Autonomous results

Each score cell is **R1 / R2 / RL F1**. These are soft set metrics: extra and missing outputs are penalized. R1 measures unigram overlap, R2 adjacent-token overlap, and RL longest-common-subsequence overlap.

MODEL	N	B	M	S	DAGER ADAPTED	FILM-INSPIRED BEAM	V2 FRAGMENTS
Pythia-160m	1	1	1	64	0.657 / 0.657 / 0.657	0.102 / 0.082 / 0.062	0.031 / 0.024 / 0.031
Pythia-160m	2	2	4	64	0.055 / 0.029 / 0.055	0.207 / 0.086 / 0.114	0.036 / 0.030 / 0.036
Pythia-160m	4	2	8	128	0.054 / 0.022 / 0.054	0.228 / 0.053 / 0.133	0.032 / 0.027 / 0.032
Pythia-160m	16	2	32	64	0.044 / 0.000 / 0.044	0.120 / 0.035 / 0.077	0.035 / 0.026 / 0.034
Pythia-410m	1	1	1	64	0.657 / 0.657 / 0.657	0.084 / 0.059 / 0.053	0.044 / 0.031 / 0.044
Pythia-410m	4	1	4	128	0.053 / 0.022 / 0.053	0.194 / 0.079 / 0.100	0.026 / 0.022 / 0.026
Pythia-410m	4	2	8	128	0.055 / 0.022 / 0.055	0.236 / 0.073 / 0.138	0.027 / 0.022 / 0.027
Pythia-410m	8	2	16	128	0.042 / 0.020 / 0.042	0.152 / 0.036 / 0.104	0.026 / 0.019 / 0.026



Exact recovery. Full exact sequence-set F1 is 0 in all 24 method/cell evaluations. Strong partial leakage is still present: singleton DAGER is nearly exact, while support recovery remains high for FILM/V2.

4. V2 truth-assisted diagnostic

V2 naturally returns fragments. The preserved mentor evaluator uses the private reference to assign fragments to sentences and order them. These values test compatibility with the mentor artifact, but are not autonomous attack scores.

MODEL	N	B	S	ASSISTED R1/R2/RL	AUTONOMOUS RL
Pythia-160m	1	1	64	0.914 / 0.794 / 0.800	0.031
Pythia-160m	2	2	64	0.798 / 0.689 / 0.780	0.036
Pythia-160m	4	2	128	0.764 / 0.631 / 0.761	0.032
Pythia-160m	16	2	64	0.612 / 0.439 / 0.602	0.034
Pythia-410m	1	1	64	0.831 / 0.667 / 0.831	0.044
Pythia-410m	4	1	128	0.779 / 0.639 / 0.752	0.026
Pythia-410m	4	2	128	0.770 / 0.637 / 0.762	0.027
Pythia-410m	8	2	128	0.718 / 0.539 / 0.701	0.026

Do not compare the assisted column as if it were another autonomous method. It has evaluator-only truth access. The fair primary V2 values are the autonomous fragment-set columns in Section 3.

5. Support, rank and resources

MODEL	M×S	METHOD	OUTPUTS	SUPPORT F1	MATCHED RL	ATTACK TIME	PEAK GPU	DAGER RANK
Pythia-160m	1×64	DAGER	2	1.000	0.986	12.1 s	1.98 GB	35
Pythia-160m	1×64	FILM	8	1.000	0.280	21.5 s	1.98 GB	—
Pythia-160m	1×64	V2	12	1.000	0.200	11.5 s	3.70 GB	—
Pythia-160m	4×64	DAGER	8	1.000	0.083	12.8 s	1.98 GB	189
Pythia-160m	4×64	FILM	8	1.000	0.170	5.8 s	1.99 GB	—
Pythia-160m	4×64	V2	49	1.000	0.238	71.1 s	3.77 GB	—
Pythia-160m	8×128	DAGER	13	0.998	0.071	15.6 s	1.99 GB	387
Pythia-160m	8×128	FILM	8	1.000	0.133	17.4 s	2.01 GB	—
Pythia-160m	8×128	V2	101	1.000	0.216	137.3 s	4.30 GB	—
Pythia-160m	32×64	DAGER	32	0.351	0.044	74.0 s	1.98 GB	767
Pythia-160m	32×64	FILM	8	1.000	0.048	26.8 s	1.99 GB	—
Pythia-160m	32×64	V2	300	1.000	0.179	415.2 s	6.14 GB	—
Pythia-410m	1×64	DAGER	2	1.000	0.986	20.7 s	4.89 GB	35
Pythia-410m	1×64	FILM	8	1.000	0.240	11.5 s	4.89 GB	—
Pythia-410m	1×64	V2	12	1.000	0.286	29.8 s	7.94 GB	—

MODEL	M×S	METHOD	OUTPUTS	SUPPORT F1	MATCHED RL	ATTACK TIME	PEAK GPU	DAGER RANK
Pythia-410m	4×128	DAGER	8	1.000	0.080	12.8 s	4.89 GB	206
Pythia-410m	4×128	FILM	8	1.000	0.151	18.9 s	4.91 GB	—
Pythia-410m	4×128	V2	56	1.000	0.198	197.2 s	8.01 GB	—
Pythia-410m	8×128	DAGER	11	0.996	0.065	33.0 s	4.89 GB	387
Pythia-410m	8×128	FILM	8	1.000	0.138	67.3 s	4.91 GB	—
Pythia-410m	8×128	V2	107	1.000	0.193	349.7 s	8.39 GB	—
Pythia-410m	16×128	DAGER	9	0.993	0.033	13.7 s	4.89 GB	781
Pythia-410m	16×128	FILM	8	1.000	0.078	33.9 s	4.91 GB	—
Pythia-410m	16×128	V2	203	1.000	0.176	654.9 s	9.07 GB	—

V2 is the runtime bottleneck: its deepest Pythia-410m cell required 654.9 seconds and 9.07 GB peak allocated GPU memory. DAGER and FILM remained substantially faster. Runtime values vary across replays; output and metric equality—not wall-clock equality—is the determinism criterion.

MODEL/CELL	DAGER BASIS	MEASURED RANK	HIDDEN WIDTH	DESIGNATION
Pythia-160m M1S64	31	35	768	below
Pythia-160m M4S64	136	189	768	below
Pythia-160m M8S128	259	387	768	below
Pythia-160m M32S64	747	767	768	boundary
Pythia-410m M1S64	31	35	1024	below
Pythia-410m M4S128	148	206	1024	below
Pythia-410m M8S128	259	387	1024	below
Pythia-410m M16S128	473	781	1024	below

Nominal M×S is not measured rank. Pythia-160m M32S64 reaches 767/768 and is designated boundary. Pythia-410m M16S128 remains measured below at rank 781/1024 despite being a nominal above-wall design.

6. Qualitative reconstruction

Private development reference (evaluator view)

As an easily measurable factor P wave duration merits further studies with higher number of patients to evaluate its importance in the estimation of stroke risk in type 2 diabetic patients with PNMMVD.

DAGER

As an easily measurable factor P wave duration merits further studies with higher number of patients to evaluate its importance in the estimation of stroke risk in type 2 diabetic patients with PNMMVD

Near-perfect: matched RL 0.986 and visible-prefix exact recall 1.000. The final word “PNMMVD” is recovered; only the final endpoint punctuation/token is absent. The second candidate is:

That repetitive extra output lowers autonomous set RL to 0.657. This explains why best-match quality and fair set quality differ.

estimation of the number of patients in the patients with the risk of stroke in patients of higher risk
with higher stroke risk in stroke patients to evaluate the importance of risk factor estimation in
further studies to further evaluate its importance in risk estimation studies in studies with studies
of its risk to the type of studies the studies evaluate in its type

- evaluate its importance
- in type 2 diabetic
- of stroke risk in the estimation
- of patients to
- further studies
- wave duration merits further

8. Limits and remaining blockers

Not yet a publication-grade final benchmark. Numerical reproducibility is strong, but the campaign still lacks held-out final rows, repeated independent victim aggregates, confidence intervals and enforced three-process victim → attacker → evaluator isolation.

- **Development-only:** first development rows, seed 0, one aggregate per cell.
- **Process boundary:** current development runner keeps victim, attacks and evaluator in one host process and passes in-memory gradient dictionaries. Restricted views exist and were hashed, but attack subprocess reopening is not yet physically enforced.
- **Evaluator boundary:** predictions are frozen and receipted, but a separate evaluator process reopening the receipt is not yet the campaign default.
- **FILM identity/provenance:** missing planned candidate-gradient-norm reorder; do not claim a complete FILM reproduction. The implementation computes a support hash in memory, but the development pilot does not persist that support hash into the artifact/receipt.
- **DAGER identity/cardinality:** GPT-NeoX fused-QKV right-subspace extension, not official Pythia support. It can emit more than public M; the set evaluator correctly penalizes this, but the adaptation's output contract needs freezing.
- **V2 oracle:** assisted scores use private truth and must never be promoted to autonomous performance.
- **No endpoint completion:** causal final target behavior remains separately visible; no truth-assisted endpoint module was added.
- **Victim provenance:** toy tests support global-mean/padding behavior, but a protocol-level real-Pythia native-PAD versus EOS-as-PAD equivalence check is not yet evidenced. The recorded model-state digest binds repository/revision/dtype metadata, not the actual weight bytes.
- **No training:** no victim/prior/tokenizer training or fine-tuning occurred.

Allowed claim: “The current implementation deterministically reproduces these eight seed-0 development results from byte-verified gradients and frozen predictions.”

Not allowed: “These are final expected benchmark accuracies” or “all three published methods have been fully reproduced.”

9. Provenance and reproducibility

REPOSITORY HEAD	106dd2e170837adc2d677d81d84dc9df6e83b1f2
EXACT V2 SOURCE	a841fb65d655cc6ea689dc1dd5a344b51f8d683ba25c43c5ff97a4a672a0fa75
DEVELOPMENT DATA	abc7590c6361d7c4a9412c2c7cedf36369fb0a14a3e543a3fdb8bdabe13ca96d · 800 rows
PYTHIA-160M REVISION	50f5173d932e8e61f858120bcb800b97af589f46
PYTHIA-410M REVISION	9879c9b5f8bea9051dcb0e68dff21493d67e9d4f
CONFIRMED-RESULTS SOURCE JSON	0f152be5047225a8c6c442d685613d4ababcf723811950ecb3c87c582f9ba97
ALL-EIGHT REPLAY PROOF	5a4f44eed1b3cd0a353797c316ce6a6ee8e516dd87dda777ea31fa34400c2a01
COMBINED EVIDENCE BUNDLE	cda2b3c36d26d7e5e0063956d1ce39661fa3ebf6a131607ef5f442fec305546b

Environment

```
{  
  "python": "3.12.3",  
  "torch": "2.8.0+cu128",  
  "transformers": "4.55.2",  
  "safetensors": "0.6.2",  
  "cuda_runtime": "12.8",  
  "gpu": [  
    "NVIDIA RTX 4000 Ada Generation",  
    "NVIDIA RTX 4000 Ada Generation"  
  ]  
}
```

Quality gates

```
pytest: 57 passed  
ruff check src tests scripts: passed  
ruff format --check src tests scripts: 35 files  
clean  
mypy src: success, 20 source files  
Git worktree: clean  
V2/data checksums: unchanged
```

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